



Aprendizagem 2025

Lab 3: Bayesian learning

Practical exercises

I.

Probability theory

	A	B	C	D
x_1	1	1	0	0
x_2	1	1	1	0
x_3	0	0	0	1
x_4	0	0	0	1
x_5	0	0	0	0
x_6	0	0	0	0

Consider the following registry where an experiment is repeated six times and four events (A, B, C and D) are detected.

Considering frequentist estimates, compute:

$$p(A)$$

$$p(A, B)$$

$$p(B|A)$$

$$p(A, B, C)$$

$$p(A|B, C)$$

$$p(A, B, C, D)$$

$$p(D|A, B, C)$$

1. Considering the following two-dimensional measurements $\{(-2,2),(-1,3),(0,1),(-2,1)\}$.
 - a) What are the maximum likelihood parameters of a multivariate Gaussian distribution for this set of points?
 - b) What is the shape of the Gaussian? Draw it approximately using a contour map.

II.

Bayesian learning

	y1	y2	y3	y4	y5	class
x1	1	1	0	1	0	1
x2	1	1	1	0	0	0
x3	0	1	1	1	0	0
x4	0	0	0	1	1	0
x5	1	0	1	1	1	1
x6	0	0	1	0	0	1
x7	0	0	0	0	1	1

Consider the following dataset where:

- 0: False and 1: True
- y1: Fast processing
- y2: Decent Battery
- y3: Good Camera
- y4: Good Look and Feel
- y5: Easiness of Use
- class: iPhone

And the query vector $\mathbf{x}_{\text{new}} = [1 \ 1 \ 1 \ 1 \ 1]^T$

- Using Bayes' rule, without making any assumptions, compute the posterior probabilities for the query vector. How is it classified?
- What is the problem of working without assumptions?
- Compute the class for the same query vector under the naive Bayes assumption.
- Consider the presence of missings. Under the same naive Bayes assumption, how do you classify $\mathbf{x}_{\text{new}} = [1 \ ? \ 1 \ ? \ 1]^T$

	weight (kg)	height (cm)	NBA player
x1	170	160	0
x2	80	220	1
x3	90	200	1
x4	60	160	0
x5	50	150	0
x6	70	190	1

Consider the following dataset

And the query vector $\mathbf{x}_{\text{new}} = [100 \ 225]^T$

- a) Compute the most probable class for the query vector assuming that the likelihoods are 2-dimensional Gaussians
- b) Compute the most probable class for the query vector, under the Naive Bayes assumption, using 1-dimensional Gaussians to model the likelihoods

5. Assuming training examples with m Boolean features.

- a) How many parameters do you have to estimate considering features are Boolean and:
 - i. no assumptions about how the data is distributed
 - ii. naive Bayes assumption
- b) How many parameters do you have to estimate considering features are numeric and:
 - iii. multivariate Gaussian assumption
 - iv. naive Bayes with Gaussian assumption

Programming quests

Resources: *Classification* and *Evaluation* notebooks available at the course's webpage

- 6. Reuse the **sklearn** code from last lab where we learnt a decision tree in the *breast.w* data:
 - a) apply the naïve Bayes classifier with default parameters
 - b) compare the accuracy of both classifiers using a 10-fold cross-validation